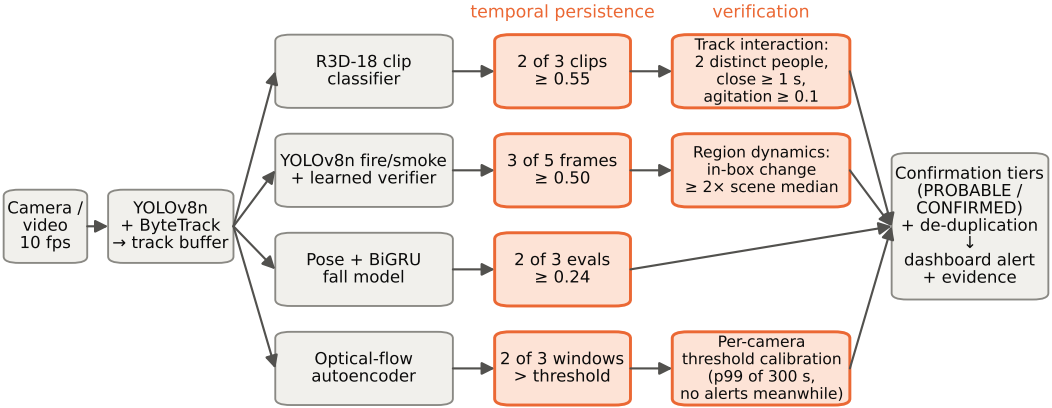
temporal persistence
verification
Camera / video 10 fps
YOLOv8n + ByteTrack → track buffer
R3D-18 clip classifier
YOLOv8n fire/smoke + learned verifier
Pose + BiGRU fall model
Optical-flow autoencoder
2 of 3 clips ≥ 0.55
3 of 5 frames ≥ 0.50
2 of 3 evals ≥ 0.24
2 of 3 windows > threshold
Track interaction: 2 distinct people, close ≥ 1 s, agitation ≥ 0.1
Region dynamics: in-box change ≥ 2× scene median
Per-camera threshold calibration (p99 of 300 s, no alerts meanwhile)
Confirmation tiers (PROBABLE / CONFIRMED) + de-duplication
↓
dashboard alert + evidence